

Medicaid Expansion and Spending by Payer

Results

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Design

Data	DEX 2.0, state $\times$ year $\times$ payer $\times$ type of care, 2010–2018
Outcome basis	<i>All ages</i> (headline); <i>Age/sex-standardized</i> shown as robustness
Treatment	First full calendar year of expansion exposure
Sample	25 states expanding in 2014 vs 19 never expanding by 2018
Estimator	Callaway–Sant’Anna, not-yet-treated controls, SEs clustered by state
Main spec	Outcome regression; poverty, income, prior uninsurance, age shares
Robustness	dr and ipw on a reduced covariate set

1,332 models estimated in total: 6 outcomes $\times$ 6 settings $\times$ 4 payers $\times$ 2 age bases $\times$ 2 samples $\times$ 3 estimators. All converge. Every estimate in this deck is the simple aggregation of group-time ATTs; event-study plots use the dynamic aggregation.

Outcomes are the terms of one accounting identity

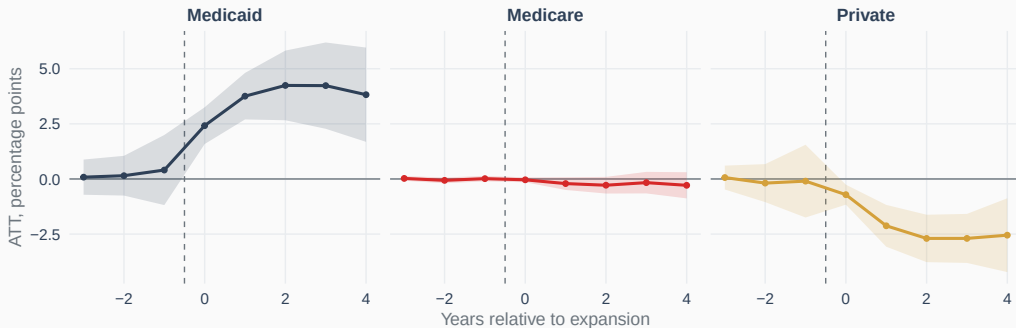
$$\underbrace{\frac{\text{spend}}{\text{pop}}}_{\text{per capita}} = \underbrace{\frac{\text{spend}}{\text{vol}}}_{\text{price}} \times \underbrace{\frac{\text{vol}}{\text{enrollees}}}_{\text{use per enrollee}} \times \underbrace{\frac{\text{enrollees}}{\text{pop}}}_{\text{coverage}}$$

Enrollees and population are recovered from the DEX ratios and reconcile exactly on the *All ages* basis. Each term is estimated separately below, so the channels can be read against one another rather than inferred.

First stage

Does expansion move coverage?

Medicaid coverage rises 3.7 points; private coverage falls 2.2



Enrollees as a percentage of state population. Main specification. Overall ATTs: Medicaid +3.69 (SE 0.71), Medicare -0.20 (0.18), private -2.16 (0.45). Pre-trend p of 0.84, 0.72 and 0.96 respectively.

The first stage across every specification

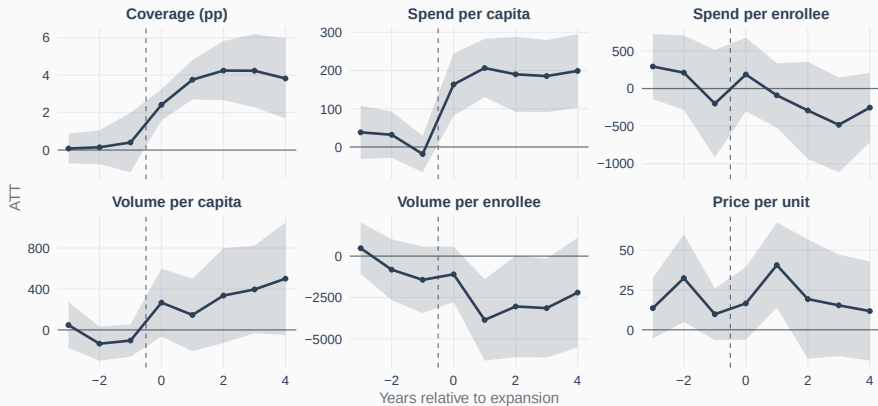
Estimator	Age basis	Sample	Medicaid	Medicare	Private
dr	all ages	all	4.2*** (0.5)	-0.1 (0.1)	-1.6*** (0.5)
dr	all ages	2014	4.2*** (0.5)	-0.1 (0.1)	-1.4*** (0.5)
dr	std	all	4.6*** (0.5)	-0.0 (0.3)	-0.8 (1.3)
dr	std	2014	4.4*** (0.6)	-0.2 (0.3)	-1.5 (1.4)
ipw	all ages	all	3.9*** (0.4)	-0.1 (0.1)	-1.8*** (0.4)
ipw	all ages	2014	3.9*** (0.5)	-0.1 (0.1)	-1.7*** (0.4)
ipw	std	all	4.3*** (0.4)	-0.1 (0.2)	-1.4 (1.3)
ipw	std	2014	4.1*** (0.5)	-0.3 (0.2)	-2.1* (1.2)
reg	all ages	all	3.7*** (0.7)	-0.2 (0.1)	-2.1*** (0.4)
reg	all ages	2014	3.7*** (0.7)	-0.2 (0.2)	-2.2*** (0.5)
reg	std	all	4.0*** (0.8)	0.0 (0.3)	-0.9 (1.2)
reg	std	2014	3.8*** (0.9)	-0.2 (0.3)	-1.8 (1.2)

Coverage in percentage points, standard errors clustered by state in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. The Medicaid and private results hold across all twelve; the private estimate loses precision on the standardized basis, where the implied denominator is not a population.

Medicaid

Which term of the identity moves?

Only coverage and per-capita spending move



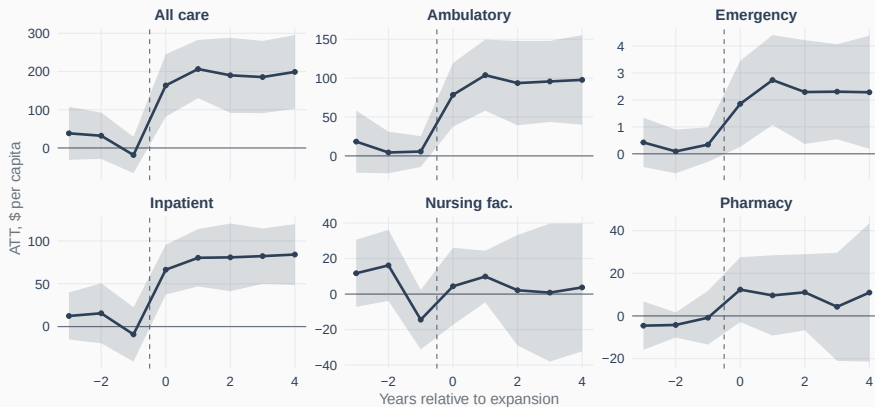
Medicaid, all care. Each panel is a separate model on its own scale.

Medicaid: all six outcomes by setting

Outcome	All care	Ambulatory	Emergency	Inpatient	Nursing fac.	Pharmacy
Coverage (pp)	3.7*** (0.7)	—	—	—	—	—
Spend per capita	188.9*** (42.6)	93.9*** (24.1)	2.3*** (0.8)	78.8*** (14.9)	4.2 (13.5)	9.7 (10.9)
Spend per enrollee	-186.5 (248.4)	0.7 (124.7)	-1.1 (4.2)	83.0 (86.2)	-163.9 (105.4)	-103.4 (78.3)
Volume per capita	328.9* (191.0)	297.5** (151.3)	15.9 (12.8)	10.6*** (2.7)	0.6 (1.0)	4.3 (163.4)
Volume per enrollee	-2,672.6** (1,252.3)	-442.4 (927.4)	-61.5 (69.5)	9.4 (19.5)	-7.9 (7.9)	-2,143.9 (1,310.2)
Price per unit	20.8 (15.1)	12.9** (5.4)	9.8*** (2.5)	505.5** (247.7)	-73.7 (139.7)	0.4 (2.2)

Overall ATTs, main specification, standard errors clustered by state. Coverage is defined per payer and does not vary by setting. Units differ by row: percentage points, dollars per capita, dollars per enrollee, units per 1,000, and dollars per unit.

Medicaid per-capita spending, by setting

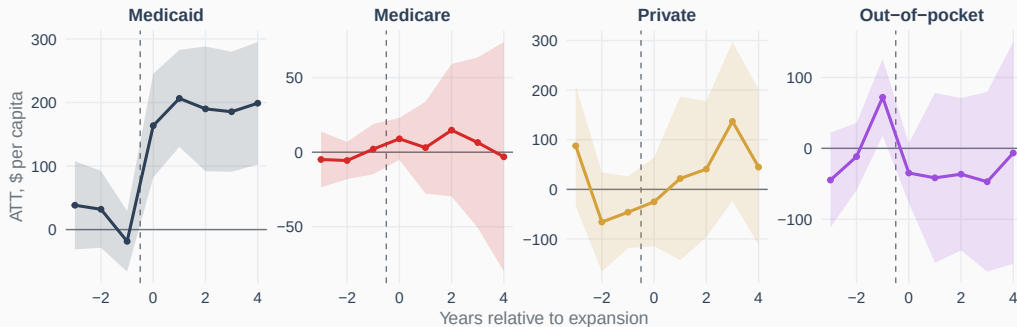


Ambulatory and inpatient carry the aggregate; nursing facility and pharmacy are flat.

All payers

Where does the money go

Per-capita spending by payer, all care



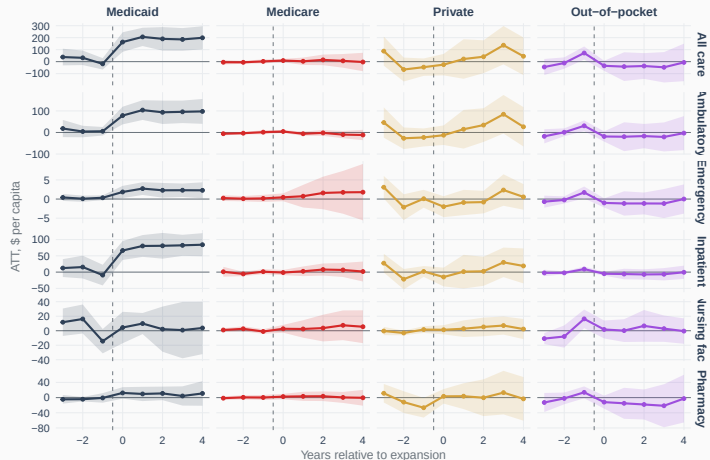
Note the free vertical scale on each panel. Private and out-of-pocket confidence bands include zero throughout the post-period.

Per-capita spending, every payer and setting

Payer	All care	Ambulatory	Emergency	Inpatient	Nursing fac.	Pharmacy
Medicaid	188.9*** (42.6)	93.9*** (24.1)	2.3*** (0.8)	78.8*** (14.9)	4.2 (13.5)	9.7 (10.9)
Medicare	6.0 (21.6)	-5.0 (6.1)	1.3 (2.1)	3.5 (8.4)	4.3 (8.3)	1.9 (5.9)
Private	44.0 (63.3)	29.5 (33.1)	-0.2 (1.3)	7.6 (21.5)	3.7 (5.7)	3.3 (20.3)
Out-of-pocket	-33.3 (54.1)	-15.7 (32.3)	-0.9 (1.5)	-5.0 (8.2)	2.2 (8.2)	-13.8 (23.2)

Dollars per state resident, main specification, standard errors clustered by state. $*p < 0.10$, $**p < 0.05$, $***p < 0.01$.

The full event-study grid

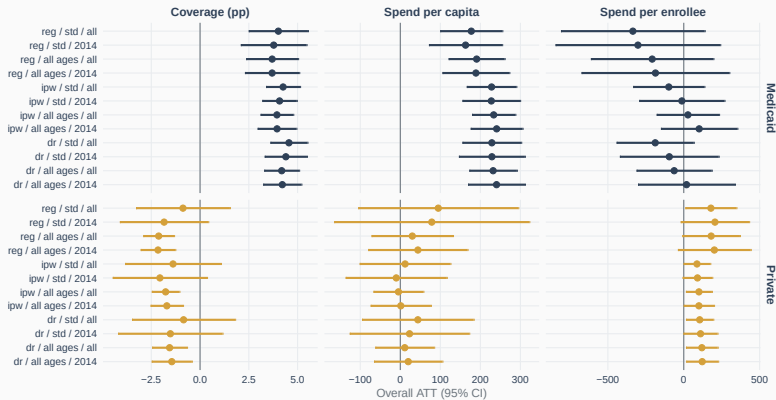


Per-capita spending, 24 models, free vertical scale within each panel.

Robustness

Estimator, age basis, sample

Headline estimates across twelve specifications



Rows are estimator / age basis / sample. Coverage and per-capita spending are stable in sign and magnitude; spend per enrollee straddles zero everywhere.

Pre-trend tests across all 1,332 models



Age basis	Sample	Models	Pass pre-test
All ages	All cohorts	333	0 (0%)
All ages	2014 cohort	333	249 (75%)
Standardized	All cohorts	333	0 (0%)
Standardized	2014 cohort	333	254 (76%)

The joint Wald test is severely oversized when treated cohorts are small, and the cohorts here are 25, 3, 3 and 1. In placebos on never-expanded states, a one-state cohort fails 88.5% of the time with no treatment. Read the 2014-cohort rows only.

What the estimates say, by term

Term	Medicaid	Private	Pre-test
Coverage (pp)	3.7*** (0.7)	-2.2*** (0.5)	both pass
Price per unit (\$)	20.8 (15.1)	5.5 (6.4)	Medicaid fails
Volume per enrollee	-2,672.6** (1,252.3)	503.1 (378.5)	private fails
Spend per enrollee (\$)	-186.5 (248.4)	201.5* (122.5)	both pass
Spend per capita (\$)	188.9*** (42.6)	44.0 (63.3)	both pass

All care, main specification, standard errors clustered by state. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

“Pre-test” reports whether the joint pre-trend test rejects at the 5% level.